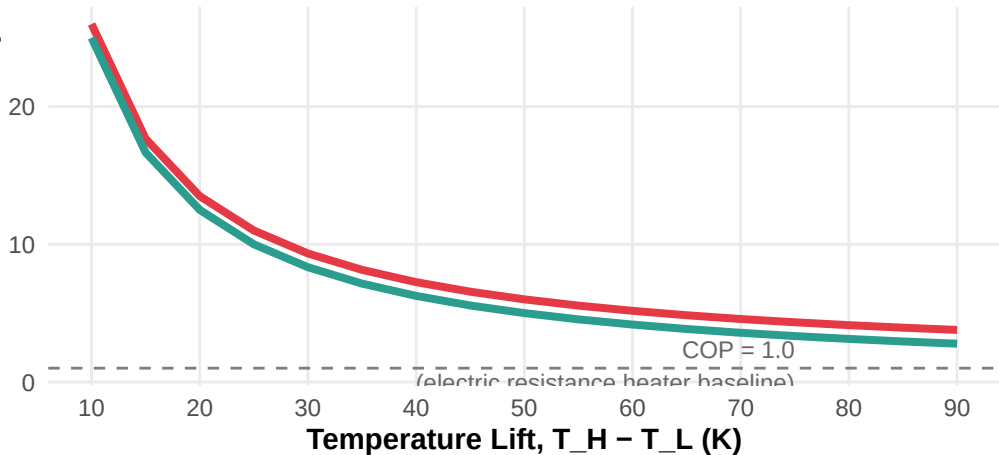


Carnot COP vs. Temperature Lift ($T_L = 250\text{ K}$)

$$\text{COP}_{\text{HP}} = \text{COP}_{\text{R}} + 1 \text{ at all conditions}$$

Coefficient of Performance (COP)



Heat Pump (COP_HP) Refrigerator (COP_R)